Kathmandu University

Assignment II – Solutions

MATH 509: Real Analysis

Text Book: Mathematical Analysis (2nd Edition)
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M.Sc. Mathematics

Contents

1 Section A: Metric Spaces and Continuity

1.1 Definitions

Definition 1.1 (Metric Space). A **metric space** is a pair (S, d) where S is a nonempty set and $d : S \times S \to \mathbb{R}$ is a function (called a **metric** or **distance function**) satisfying for all $x, y, z \in S$:

(M1) $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ (positive definiteness).

(M2) $d(x, y) = d(y, x)$ (symmetry).

(M3) $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).

Definition 1.2 (Convergent Sequence in a Metric Space). Let (S, d) be a metric space. A sequence $\{p_n\}$ in S is said to **converge** to a point $p \in S$ if for every $\varepsilon > 0$ there exists an integer N such that

$$d(p_n, p) < \varepsilon \quad \text{for all } n \geq N.$$

We write $p_n \to p$ or $\lim_{n\to\infty} p_n = p$.

Definition 1.3 (Cauchy Sequence in a Metric Space). A sequence $\{p_n\}$ in a metric space (S, d) is called a **Cauchy sequence** if for every $\varepsilon > 0$ there exists an integer N such that

$$d(p_m, p_n) < \varepsilon \quad \text{for all } m, n \geq N.$$

Definition 1.4 (Complete Metric Space). A metric space (S, d) is said to be **complete** if every Cauchy sequence in S converges to a point in S. Examples: $\mathbb{R}^n$ with the Euclidean metric is complete; $\mathbb{Q}$ with the usual metric is not complete.

Definition 1.5 (Continuous Function in a Metric Space). Let (S, d_S) and (T, d_T) be metric spaces. A function $f : S \to T$ is **continuous at a point** $p \in S$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_T(f(x), f(p)) < \varepsilon \quad \text{whenever } d_S(x, p) < \delta.$$

The function f is **continuous on** S if it is continuous at every point of S.

Definition 1.6 (Uniformly Continuous Function in a Metric Space). Let (S, d_S) and (T, d_T) be metric spaces. A function $f : S \to T$ is **uniformly continuous on** S if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_T(f(x), f(y)) < \varepsilon \quad \text{for all } x, y \in S \text{ with } d_S(x, y) < \delta.$$

Note: The crucial distinction from pointwise continuity is that δ depends only on ε, not on any particular point.

1.2 Theorems

Theorem 1.1 (Theorem 3.34 – Convergent Sequences are Cauchy). In a metric space (S, d), every convergent sequence is a Cauchy sequence.

Proof. Let $\{p_n\}$ converge to p in S. Let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that $d(p_n, p) < \varepsilon/2$ for all $n \geq N$.

For $m, n \geq N$, by the triangle inequality:

$$d(p_m, p_n) \leq d(p_m, p) + d(p, p_n) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $\{p_n\}$ is a Cauchy sequence. □

Theorem 1.2 (Theorem 3.38 – Cauchy Sequence with Convergent Subsequence)**.** In a metric space (S, d), if a Cauchy sequence $\{p_n\}$ has a subsequence $\{p_{n_k}\}$ that converges to a point $p \in S$, then the entire sequence converges to p.

Proof. Let $\varepsilon > 0$.

Since $\{p_n\}$ is Cauchy, there exists $N_1 \in \mathbb{N}$ such that

$$d(p_m, p_n) < \frac{\varepsilon}{2} \quad \text{for all } m, n \geq N_1.$$

Since $p_{n_k} \to p$, there exists $K \in \mathbb{N}$ such that

$$d(p_{n_k}, p) < \frac{\varepsilon}{2} \quad \text{for all } k \geq K.$$

Choose $k_0 \geq K$ such that $n_{k_0} \geq N_1$. Then for all $n \geq N_1$:

$$d(p_n, p) \leq d(p_n, p_{n_{k_0}}) + d(p_{n_{k_0}}, p) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $p_n \to p$. □

Theorem 1.3 (Theorem 4.2 – Sequential Criterion for Limits of Functions)**.** Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, $f : A \to T$, and let p be an accumulation point of A. Then

$$\lim_{x \to p} f(x) = q$$

if and only if for every sequence $\{p_n\}$ in A with $p_n \neq p$ for all n and $p_n \to p$, we have $f(p_n) \to q$.

Proof. ($\Rightarrow$): Suppose $\lim_{x \to p} f(x) = q$. Let $\{p_n\}$ be a sequence in A with $p_n \neq p$ and $p_n \to p$. Let $\varepsilon > 0$. There exists $\delta > 0$ such that

$$d_T(f(x), q) < \varepsilon \quad \text{whenever } x \in A \text{ and } 0 < d_S(x, p) < \delta.$$

Since $p_n \to p$, there exists N with $d_S(p_n, p) < \delta$ for all $n \geq N$. Since $p_n \neq p$, we have $0 < d_S(p_n, p) < \delta$, so $d_T(f(p_n), q) < \varepsilon$ for all $n \geq N$. Hence $f(p_n) \to q$.

($\Leftarrow$): We prove the contrapositive. Suppose $\lim_{x \to p} f(x) \neq q$. Then there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$, there exists $p_n \in A$ with $0 < d_S(p_n, p) < 1/n$ but $d_T(f(p_n), q) \geq \varepsilon_0$.

Then $\{p_n\}$ is a sequence in A with $p_n \neq p$, $p_n \to p$, but $f(p_n) \not\to q$. This shows the condition fails. □

Theorem 1.4 (Theorem 4.8 – Open Set Characterization of Continuity)**.** Let (S, d_S) and (T, d_T) be metric spaces. A function $f : S \to T$ is continuous on S if and only if $f^{-1}(G)$ is open in S for every open set G in T.

Proof. ($\Rightarrow$): Suppose f is continuous on S. Let G be open in T. If $f^{-1}(G) = \emptyset$, it is open. Otherwise, let $p \in f^{-1}(G)$, so $f(p) \in G$. Since G is open, there exists $\varepsilon > 0$ with $B_T(f(p), \varepsilon) \subseteq G$.

Since f is continuous at p, there exists $\delta > 0$ such that $d_T(f(x), f(p)) < \varepsilon$ whenever $d_S(x, p) < \delta$. Thus $f(B_S(p, \delta)) \subseteq B_T(f(p), \varepsilon) \subseteq G$, so $B_S(p, \delta) \subseteq f^{-1}(G)$. Hence p is an interior point, and $f^{-1}(G)$ is open.

($\Leftarrow$): Suppose $f^{-1}(G)$ is open for every open $G \subseteq T$. Let $p \in S$ and $\varepsilon > 0$. The ball $B_T(f(p), \varepsilon)$ is open in T, so $f^{-1}(B_T(f(p), \varepsilon))$ is open in S.

Since $p \in f^{-1}(B_T(f(p), \varepsilon))$ and this set is open, there exists $\delta > 0$ with $B_S(p, \delta) \subseteq f^{-1}(B_T(f(p), \varepsilon))$. Therefore $d_S(x, p) < \delta \implies d_T(f(x), f(p)) < \varepsilon$. Hence f is continuous at p. □

Theorem 1.5 (Theorem 4.10 – Continuous Image of a Compact Set)**.** Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be continuous. If $K \subseteq S$ is compact, then $f(K)$ is compact in T.

Proof. Let $\{G_\alpha\}_{\alpha \in I}$ be an open cover of $f(K)$: $f(K) \subseteq \bigcup_\alpha G_\alpha$.

Since f is continuous, each $f^{-1}(G_\alpha)$ is open in S by Theorem 4.8. Moreover:

$$K \subseteq f^{-1}(f(K)) \subseteq f^{-1}\left(\bigcup_\alpha G_\alpha\right) = \bigcup_\alpha f^{-1}(G_\alpha).$$

So $\{f^{-1}(G_\alpha)\}_\alpha$ is an open cover of K. Since K is compact, there exist $\alpha_1, \ldots, \alpha_n$ such that

$$K \subseteq f^{-1}(G_{\alpha_1}) \cup \cdots \cup f^{-1}(G_{\alpha_n}).$$

Applying f:

$$f(K) \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}.$$

Hence $f(K)$ is compact. □

Theorem 1.6 (Theorem 4.17 – Uniform Continuity on Compact Metric Spaces)**.** If (S, d_S) is a compact metric space and $f : S \to T$ is continuous, then f is uniformly continuous on S.

Proof. Suppose, for contradiction, that f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ and sequences $\{x_n\}, \{y_n\}$ in S such that

$$d_S(x_n, y_n) < \frac{1}{n} \quad \text{but} \quad d_T(f(x_n), f(y_n)) \geq \varepsilon_0 \quad \text{for all } n.$$

Since S is compact (hence sequentially compact), $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ converging to some $p \in S$.

Since $d_S(x_{n_k}, y_{n_k}) < 1/n_k \to 0$ and $x_{n_k} \to p$, we also have $y_{n_k} \to p$.

By continuity of f at p: $f(x_{n_k}) \to f(p)$ and $f(y_{n_k}) \to f(p)$.

Therefore $d_T(f(x_{n_k}), f(y_{n_k})) \to 0$, contradicting $d_T(f(x_{n_k}), f(y_{n_k})) \geq \varepsilon_0$.

Hence f is uniformly continuous on S. □

Theorem 1.7 (Theorem 4.22 – Extreme Value Theorem in Metric Spaces)**.** Let (S, d) be a compact metric space and $f : S \to \mathbb{R}$ be continuous. Then f is bounded and attains its supremum and infimum on S.

Proof. By Theorem 4.10, $f(S)$ is a compact subset of $\mathbb{R}$. By the Heine-Borel theorem, compact subsets of $\mathbb{R}$ are closed and bounded.

Bounded: Since $f(S)$ is bounded, f is bounded on S.

Attains supremum: Let $M = \sup f(S)$. Since $f(S)$ is bounded, M is finite. For each $n \in \mathbb{N}$, $M - 1/n$ is not an upper bound of $f(S)$, so there exists $x_n \in S$ with $f(x_n) > M - 1/n$. Thus $f(x_n) \to M$.

Since S is compact (sequentially compact), $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ with $x_{n_k} \to p \in S$. By continuity, $f(x_{n_k}) \to f(p)$. But also $f(x_{n_k}) \to M$, so $f(p) = M$.

Attains infimum: Similarly, there exists $q \in S$ with $f(q) = \inf f(S)$. □

Theorem 1.8 (Theorem 4.23 – Preservation of Connectedness)**.** Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be continuous. If S is connected, then $f(S)$ is connected.

Proof. Suppose, for contradiction, that $f(S)$ is disconnected. Then there exist two nonempty sets A, B that are open in $f(S)$ such that:

$$f(S) = A \cup B, \quad A \cap B = \emptyset, \quad A \neq \emptyset, \quad B \neq \emptyset.$$

Let $U = f^{-1}(A)$ and $V = f^{-1}(B)$. Since there exist open sets G_1, G_2 in T with $A = f(S) \cap G_1$ and $B = f(S) \cap G_2$, we have $U = f^{-1}(G_1)$ and $V = f^{-1}(G_2)$, which are open in S (by continuity).

Moreover:

- $U \cup V = f^{-1}(A) \cup f^{-1}(B) = f^{-1}(A \cup B) = f^{-1}(f(S)) = S$.
- $U \cap V = f^{-1}(A) \cap f^{-1}(B) = f^{-1}(A \cap B) = f^{-1}(\emptyset) = \emptyset$.
- $U \neq \emptyset$ since $A \neq \emptyset$, and $V \neq \emptyset$ since $B \neq \emptyset$.

This gives a disconnection of S, contradicting the assumption that S is connected. Hence $f(S)$ is connected. □

Corollary 1.9 (Intermediate Value Theorem)**.** If $f : [a, b] \to \mathbb{R}$ is continuous, then f takes every value between $f(a)$ and $f(b)$. That is, if $f(a) < \gamma < f(b)$ (or $f(b) < \gamma < f(a)$), there exists $c \in (a, b)$ with $f(c) = \gamma$.

Proof. The interval $[a, b]$ is connected, so $f([a, b])$ is connected by Theorem 4.23. A connected subset of $\mathbb{R}$ is an interval. Since $f(a)$ and $f(b)$ belong to $f([a, b])$, every value between them also belongs to $f([a, b])$. □

2 Section B: Functions of Bounded Variation

2.1 Definitions

Definition 2.1 (Partition)**.** A **partition** of an interval $[a, b]$ is a finite set of points $P = \{x_0, x_1, \ldots, x_n\}$ such that

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

A partition P^* is called a **refinement** of P if $P \subseteq P^*$. The **common refinement** of two partitions P_1 and P_2 is $P_1 \cup P_2$.

Definition 2.2 (Functions of Bounded Variation)**.** Let $f : [a, b] \to \mathbb{R}$. For a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$, define

$$V(P, f) := \sum_{k=1}^{n} |f(x_k) - f(x_{k-1})|.$$

The function f is said to be of **bounded variation** on $[a, b]$, written $f \in BV[a, b]$, if the set $\{V(P, f) : P \text{ is a partition of } [a, b]\}$ is bounded above.

Definition 2.3 (Total Variation)**.** If $f \in BV[a, b]$, the **total variation** of f on $[a, b]$ is

$$V_f(a, b) := \sup\{V(P, f) : P \text{ is a partition of } [a, b]\}.$$

The **total variation function** $V : [a, b] \to \mathbb{R}$ is defined by $V(a) = 0$ and $V(x) = V_f(a, x)$ for $x \in (a, b]$.

2.2 Theorems

Theorem 2.1 (Theorem 6.2 – Monotone Functions are of Bounded Variation)**.** If f is monotone on $[a, b]$, then $f \in BV[a, b]$ and $V_f(a, b) = |f(b) - f(a)|$.

Proof. Assume f is increasing (the decreasing case is analogous). For any partition $P = \{x_0, x_1, \ldots, x_n\}$:

$$V(P, f) = \sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| = \sum_{k=1}^{n} \big(f(x_k) - f(x_{k-1})\big) = f(b) - f(a),$$

since the sum telescopes. Thus $V(P, f) = f(b) - f(a)$ for *every* partition, so

$$V_f(a, b) = \sup_P V(P, f) = f(b) - f(a) = |f(b) - f(a)|.$$

For f decreasing: $|f(x_k) - f(x_{k-1})| = f(x_{k-1}) - f(x_k)$, telescoping to $f(a) - f(b) = |f(b) - f(a)|$. □

Theorem 2.2 (Theorem 6.6 – Additivity of Total Variation)**.** If $c \in (a, b)$, then $f \in BV[a, b]$ if and only if $f \in BV[a, c]$ and $f \in BV[c, b]$. Moreover,

$$V_f(a, b) = V_f(a, c) + V_f(c, b).$$

Proof. **($\leq$ direction):** Let P be any partition of $[a,b]$. Let $P^* = P \cup \{c\}$. Adding a point does not decrease variation (triangle inequality), so $V(P,f) \leq V(P^*,f)$.

Write $P^* = P_1 \cup P_2$ where $P_1 = P^* \cap [a,c]$ is a partition of $[a,c]$ and $P_2 = P^* \cap [c,b]$ is a partition of $[c,b]$. Then

$$V(P,f) \leq V(P^*,f) = V(P_1,f) + V(P_2,f) \leq V_f(a,c) + V_f(c,b).$$

Taking supremum over P: $V_f(a,b) \leq V_f(a,c) + V_f(c,b)$.

($\geq$ direction): Let P_1 be any partition of $[a,c]$ and P_2 any partition of $[c,b]$. Then $P = P_1 \cup P_2$ is a partition of $[a,b]$ (containing c), and

$$V(P_1,f) + V(P_2,f) = V(P,f) \leq V_f(a,b).$$

Taking sup over P_1: $V_f(a,c) + V(P_2,f) \leq V_f(a,b)$. Taking sup over P_2:

$$V_f(a,c) + V_f(c,b) \leq V_f(a,b).$$

Combining: $V_f(a,b) = V_f(a,c) + V_f(c,b)$. The equivalence of $f \in BV[a,b]$ with $f \in BV[a,c] \cap BV[c,b]$ follows immediately. □

Theorem 2.3 (Theorem 6.9 – Properties of Total Variation Function)**.** If $f \in BV[a,b]$, then:

(a) The total variation function $V(x) = V_f(a,x)$ is increasing on $[a,b]$.

(b) The function $V - f$ is also increasing on $[a,b]$.

Proof. **(a)** For $a \leq x < y \leq b$, by additivity (Theorem 6.6):

$$V(y) = V_f(a,y) = V_f(a,x) + V_f(x,y) = V(x) + V_f(x,y).$$

Since $V_f(x,y) \geq 0$, we have $V(y) \geq V(x)$, so V is increasing.

(b) For $a \leq x < y \leq b$, we need $(V-f)(y) \geq (V-f)(x)$, i.e., $V(y) - V(x) \geq f(y) - f(x)$.

By additivity, $V(y) - V(x) = V_f(x,y)$. Using the partition $\{x,y\}$ of $[x,y]$:

$$V_f(x,y) \geq |f(y) - f(x)| \geq f(y) - f(x).$$

Hence $V(y) - V(x) \geq f(y) - f(x)$, proving $V - f$ is increasing. □

Theorem 2.4 (Theorem 6.11 – Jordan Decomposition Theorem)**.** A function $f : [a,b] \to \mathbb{R}$ is of bounded variation on $[a,b]$ if and only if f can be written as the difference of two increasing functions.

Proof. ($\Rightarrow$): Let $f \in BV[a,b]$ and define $V(x) = V_f(a,x)$. By Theorem 6.9, both V and $g := V - f$ are increasing on $[a,b]$. Therefore

$$f(x) = V(x) - g(x)$$

expresses f as the difference of two increasing functions.

($\Leftarrow$): Suppose $f = g - h$ where g and h are increasing on $[a, b]$. For any partition $P = \{x_0, \dots, x_n\}$:

$$\begin{aligned} V(P, f) &= \sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| \\ &= \sum_{k=1}^{n} |(g(x_k) - g(x_{k-1})) - (h(x_k) - h(x_{k-1}))| \\ &\leq \sum_{k=1}^{n} |g(x_k) - g(x_{k-1})| + \sum_{k=1}^{n} |h(x_k) - h(x_{k-1})| \\ &= \sum_{k=1}^{n} (g(x_k) - g(x_{k-1})) + \sum_{k=1}^{n} (h(x_k) - h(x_{k-1})) \\ &= (g(b) - g(a)) + (h(b) - h(a)) < \infty. \end{aligned}$$

So $V(P, f)$ is bounded above for all P, hence $f \in BV[a, b]$ and $V_f(a, b) \leq (g(b) - g(a)) + (h(b) - h(a))$. □

Theorem 2.5 (Theorem 6.12 – Continuity of Total Variation Function)**.** If $f \in BV[a, b]$ and f is continuous at a point $c \in [a, b]$, then the total variation function $V(x) = V_f(a, x)$ is also continuous at c.

Proof. Since V is increasing (Theorem 6.9), the one-sided limits $V(c^+)$ and $V(c^-)$ exist. We show $V(c^+) = V(c) = V(c^-)$.

Right-continuity at c (assuming $c < b$): We need to show $\lim_{h \to 0^+} V_f(c, c+h) = 0$.

Let $\varepsilon > 0$. Since $f \in BV[c, b]$, there exists a partition $Q = \{c = t_0, t_1, \dots, t_m = b\}$ of $[c, b]$ such that

$$V(Q, f) > V_f(c, b) - \varepsilon.$$

Let $\delta = t_1 - c > 0$. Since f is continuous at c, we may also choose δ small enough that $|f(x) - f(c)| < \varepsilon$ for $|x - c| < \delta$.

For $0 < h < \delta$, the points $\{c + h, t_1, t_2, \dots, t_m\}$ form a partition of $[c + h, b]$, giving:

$$V_f(c + h, b) \geq |f(t_1) - f(c + h)| + \sum_{k=2}^{m} |f(t_k) - f(t_{k-1})|.$$

Now $V(Q, f) = |f(t_1) - f(c)| + \sum_{k=2}^{m} |f(t_k) - f(t_{k-1})|$, so:

$$V_f(c + h, b) \geq V(Q, f) - |f(t_1) - f(c)| + |f(t_1) - f(c + h)|.$$

By the reverse triangle inequality: $|f(t_1) - f(c + h)| \geq |f(t_1) - f(c)| - |f(c) - f(c + h)|$. Therefore:

$$V_f(c + h, b) \geq V(Q, f) - |f(c + h) - f(c)| > V_f(c, b) - \varepsilon - \varepsilon = V_f(c, b) - 2\varepsilon.$$

By additivity: $V_f(c, c + h) = V_f(c, b) - V_f(c + h, b) < 2\varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $V_f(c, c + h) \to 0$ as $h \to 0^+$, i.e., $V(c^+) = V(c)$.

Left-continuity at c (assuming $c > a$): We show $V_f(c - h, c) \to 0$ as $h \to 0^+$.

Let $\varepsilon > 0$. Choose a partition $Q = \{a = t_0, \dots, t_m = c\}$ of $[a, c]$ with $V(Q, f) > V_f(a, c) - \varepsilon$. Let $\delta = c - t_{m-1}$, also chosen so that $|f(x) - f(c)| < \varepsilon$ for $|x - c| < \delta$.

For $0 < h < \delta$, the partition $\{t_0, \dots, t_{m-1}, c-h\}$ of $[a, c-h]$ gives:

$$V_f(a, c-h) \geq \sum_{k=1}^{m-1} |f(t_k) - f(t_{k-1})| + |f(c-h) - f(t_{m-1})|.$$

Since $|f(c-h) - f(t_{m-1})| \geq |f(c) - f(t_{m-1})| - |f(c) - f(c-h)|$:

$$V_f(a, c-h) \geq V(Q, f) - |f(c) - f(c-h)| > V_f(a, c) - \varepsilon - \varepsilon.$$

Therefore $V_f(c-h, c) = V_f(a, c) - V_f(a, c-h) < 2\varepsilon$. Hence $V(c^-) = V(c)$. $\square$

3 Section C: Riemann-Stieltjes Integral and Sequences of Functions

3.1 Definitions

Definition 3.1 (Riemann-Stieltjes Sum)**.** Let f and α be defined on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. For each subinterval $[x_{k-1}, x_k]$, choose a point $t_k \in [x_{k-1}, x_k]$. The **Riemann-Stieltjes sum** of f with respect to α corresponding to P and the choice of points $\{t_k\}$ is

$$S(P, f, \alpha) := \sum_{k=1}^{n} f(t_k)\,[\alpha(x_k) - \alpha(x_{k-1})] = \sum_{k=1}^{n} f(t_k)\,\Delta\alpha_k.$$

We say f is **Riemann-Stieltjes integrable** with respect to α on $[a, b]$, written $f \in \mathcal{R}(\alpha)$ on $[a, b]$, if there exists $A \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists a partition P_ε with

$$|S(P, f, \alpha) - A| < \varepsilon$$

for every refinement P of P_ε and every choice of t_k. We write $A = \int_a^b f\,d\alpha$.

Definition 3.2 (Step Function)**.** A function $\alpha : [a, b] \to \mathbb{R}$ is called a **step function** if there exists a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that α is constant on each open subinterval (x_{k-1}, x_k) for $k = 1, 2, \ldots, n$.

Definition 3.3 (Sequences of Functions)**.** A **sequence of functions** on a set S is a sequence $\{f_n\}_{n=1}^{\infty}$ where each $f_n : S \to \mathbb{R}$ (or $\mathbb{R}^k$) is a function defined on S.

Definition 3.4 (Pointwise Convergence)**.** A sequence of functions $\{f_n\}$ defined on S converges **pointwise** to a function f on S if for each $x \in S$,

$$\lim_{n\to\infty} f_n(x) = f(x).$$

That is, for each $x \in S$ and each $\varepsilon > 0$, there exists $N = N(x, \varepsilon) \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$.

Definition 3.5 (Uniform Convergence of a Sequence of Functions)**.** A sequence of functions $\{f_n\}$ defined on S converges **uniformly** to f on S if for every $\varepsilon > 0$ there exists $N = N(\varepsilon) \in \mathbb{N}$ (independent of x) such that

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } n \geq N \text{ and all } x \in S.$$

Equivalently, $\sup_{x\in S} |f_n(x) - f(x)| \to 0$ as $n \to \infty$.

Definition 3.6 (Uniform Convergence of a Series of Functions)**.** A series of functions $\sum_{n=1}^{\infty} g_n$ converges **uniformly** to f on S if the sequence of partial sums $s_k(x) = \sum_{n=1}^{k} g_n(x)$ converges uniformly to f on S. That is, for every $\varepsilon > 0$ there exists N such that

$$\left| f(x) - \sum_{n=1}^{k} g_n(x) \right| < \varepsilon \quad \text{for all } k \geq N \text{ and all } x \in S.$$

3.2 Theorems

Theorem 3.1 (Theorem 7.2 – Riemann's Condition for Integrability)**.** Let α be increasing on $[a,b]$. Define the upper and lower Darboux-Stieltjes sums:

$$U(P,f,\alpha)=\sum_{k=1}^{n} M_k\,\Delta\alpha_k,\quad L(P,f,\alpha)=\sum_{k=1}^{n} m_k\,\Delta\alpha_k,$$

where $M_k=\sup_{[x_{k-1},x_k]} f$ and $m_k=\inf_{[x_{k-1},x_k]} f$.

Then $f\in\mathcal{R}(\alpha)$ on $[a,b]$ if and only if for every $\varepsilon>0$ there exists a partition P of $[a,b]$ such that

$$U(P,f,\alpha)-L(P,f,\alpha)<\varepsilon.$$

Proof. Define $\overline{\int_a^b} f\,d\alpha=\inf_P U(P,f,\alpha)$ and $\underline{\int_a^b} f\,d\alpha=\sup_P L(P,f,\alpha)$.

We always have $L(P_1,f,\alpha)\le U(P_2,f,\alpha)$ for any two partitions P_1,P_2 (by considering their common refinement). Hence $\underline{\int} f\,d\alpha\le\overline{\int} f\,d\alpha$.

($\Leftarrow$): Suppose for every $\varepsilon>0$ there exists P with $U(P,f,\alpha)-L(P,f,\alpha)<\varepsilon$. Then:

$$0\le\overline{\int_a^b} f\,d\alpha-\underline{\int_a^b} f\,d\alpha\le U(P,f,\alpha)-L(P,f,\alpha)<\varepsilon.$$

Since ε is arbitrary, $\overline{\int} f\,d\alpha=\underline{\int} f\,d\alpha$. Denote this common value by A.

For any refinement P^* of P and any Riemann-Stieltjes sum $S(P^*,f,\alpha)$:

$$L(P^*,f,\alpha)\le S(P^*,f,\alpha)\le U(P^*,f,\alpha).$$

Since P^* refines P: $U(P^*,f,\alpha)\le U(P,f,\alpha)$ and $L(P^*,f,\alpha)\ge L(P,f,\alpha)$. Also $L(P,f,\alpha)\le A\le U(P,f,\alpha)$. Therefore:

$$|S(P^*,f,\alpha)-A|\le U(P,f,\alpha)-L(P,f,\alpha)<\varepsilon.$$

Hence $f\in\mathcal{R}(\alpha)$ with $\int_a^b f\,d\alpha=A$.

($\Rightarrow$): Suppose $f\in\mathcal{R}(\alpha)$ with integral A. Let $\varepsilon>0$. There exists a partition P such that $|S(P,f,\alpha)-A|<\varepsilon/3$ for all choices of t_k.

In particular, choosing t_k to make $f(t_k)$ close to M_k gives S close to U, and choosing t_k close to m_k gives S close to L. Precisely: for any $\eta>0$, we can choose t_k with $f(t_k)>M_k-\eta$, yielding $S(P,f,\alpha)>U(P,f,\alpha)-\eta(\alpha(b)-\alpha(a))$. Letting $\eta\to 0$:

$$U(P,f,\alpha)\le A+\varepsilon/3\quad\text{and}\quad L(P,f,\alpha)\ge A-\varepsilon/3.$$

Therefore $U(P,f,\alpha)-L(P,f,\alpha)<2\varepsilon/3<\varepsilon$. □

Theorem 3.2 (Theorem 7.3 – Continuous Functions are Riemann-Stieltjes Integrable)**.** If f is continuous on $[a,b]$ and α is increasing on $[a,b]$, then $f\in\mathcal{R}(\alpha)$ on $[a,b]$.

Proof. If $\alpha(a)=\alpha(b)$, then $\Delta\alpha_k=0$ for all k in any partition, so $U=L=0$ and $f\in\mathcal{R}(\alpha)$ trivially.

Assume $\alpha(a)<\alpha(b)$. Since f is continuous on the compact set $[a,b]$, f is uniformly continuous: for every $\varepsilon>0$ there exists $\delta>0$ such that

$$|f(x)-f(y)|<\frac{\varepsilon}{\alpha(b)-\alpha(a)}\quad\text{whenever } |x-y|<\delta.$$

Choose a partition $P = \{x_0, \dots, x_n\}$ with mesh $\|P\| := \max_k(x_k - x_{k-1}) < \delta$. On each $[x_{k-1}, x_k]$, since f is continuous on a compact interval, f attains its sup M_k and inf m_k. Since the interval has length $< \delta$:

$$M_k - m_k < \frac{\varepsilon}{\alpha(b) - \alpha(a)}.$$

Therefore:

$$U(P, f, \alpha) - L(P, f, \alpha) = \sum_{k=1}^{n} (M_k - m_k)\, \Delta\alpha_k < \frac{\varepsilon}{\alpha(b) - \alpha(a)} \sum_{k=1}^{n} \Delta\alpha_k = \varepsilon.$$

By Theorem 7.2, $f \in \mathcal{R}(\alpha)$. □

Theorem 3.3 (Theorem 9.2 – Cauchy Criterion for Uniform Convergence)**.** A sequence of functions $\{f_n\}$ defined on S converges uniformly on S if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|f_m(x) - f_n(x)| < \varepsilon \quad \text{for all } m, n \geq N \text{ and all } x \in S.$$

Proof. ($\Rightarrow$): Suppose $f_n \to f$ uniformly on S. Let $\varepsilon > 0$. There exists N with $|f_n(x) - f(x)| < \varepsilon/2$ for all $n \geq N$ and $x \in S$. For $m, n \geq N$:

$$|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f(x) - f_n(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

($\Leftarrow$): Suppose the Cauchy condition holds. For each fixed $x \in S$, $\{f_n(x)\}$ is a Cauchy sequence in $\mathbb{R}$, hence converges. Define $f(x) = \lim_{n\to\infty} f_n(x)$.

We show $f_n \to f$ uniformly. Let $\varepsilon > 0$. By hypothesis, there exists N with $|f_m(x) - f_n(x)| < \varepsilon/2$ for all $m, n \geq N$ and $x \in S$.

Fix $n \geq N$ and $x \in S$. Letting $m \to \infty$:

$$|f(x) - f_n(x)| = \lim_{m\to\infty} |f_m(x) - f_n(x)| \leq \frac{\varepsilon}{2} < \varepsilon.$$

Since this holds for all $x \in S$ and $n \geq N$, the convergence is uniform. □

Theorem 3.4 (Theorem 9.3 – Uniform Limit of Continuous Functions is Continuous)**.** Let $\{f_n\}$ be a sequence of functions continuous on a metric space S. If $f_n \to f$ uniformly on S, then f is continuous on S.

Proof. Let $p \in S$ and $\varepsilon > 0$. Since $f_n \to f$ uniformly, there exists N such that

$$|f_n(x) - f(x)| < \frac{\varepsilon}{3} \quad \text{for all } n \geq N \text{ and all } x \in S.$$

Fix $n = N$. Since f_N is continuous at p, there exists $\delta > 0$ such that

$$|f_N(x) - f_N(p)| < \frac{\varepsilon}{3} \quad \text{whenever } d(x, p) < \delta.$$

For $d(x, p) < \delta$:

$$\begin{aligned} |f(x) - f(p)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(p)| + |f_N(p) - f(p)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence f is continuous at p. Since p was arbitrary, f is continuous on S. □

Theorem 3.5 (Theorem 9.5 – Interchange of Limit and Integral)**.** Let α be of bounded variation on $[a,b]$. Suppose $\{f_n\}$ is a sequence of functions with $f_n \in \mathcal{R}(\alpha)$ on $[a,b]$ for each n, and suppose $f_n \to f$ uniformly on $[a,b]$. Then $f \in \mathcal{R}(\alpha)$ on $[a,b]$ and

$$\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f_n\,d\alpha.$$

Proof. Let $V = V_\alpha(a,b)$ be the total variation of α on $[a,b]$.

Step 1: $f \in \mathcal{R}(\alpha)$**.**

Let $\varepsilon > 0$. Since $f_n \to f$ uniformly, there exists N with $|f_n(x) - f(x)| < \varepsilon/(3V+3)$ for all $n \geq N$ and $x \in [a,b]$.

Fix $n = N$. Since $f_N \in \mathcal{R}(\alpha)$, by Theorem 7.2, there exists a partition P such that $U(P, f_N, \alpha) - L(P, f_N, \alpha) < \varepsilon/3$.

Let $\eta = \varepsilon/(3V+3)$. For each subinterval $[x_{k-1}, x_k]$:

$$\sup_{[x_{k-1},x_k]} f \leq \sup_{[x_{k-1},x_k]} f_N + \eta, \qquad \inf_{[x_{k-1},x_k]} f \geq \inf_{[x_{k-1},x_k]} f_N - \eta.$$

So $M_k(f) - m_k(f) \leq (M_k(f_N) - m_k(f_N)) + 2\eta$. Therefore:

$$\begin{aligned} U(P,f,\alpha) - L(P,f,\alpha) &\leq (U(P,f_N,\alpha) - L(P,f_N,\alpha)) + 2\eta \sum |\Delta\alpha_k| \\ &< \frac{\varepsilon}{3} + 2\eta V \leq \frac{\varepsilon}{3} + \frac{2\varepsilon}{3} = \varepsilon. \end{aligned}$$

(Here we used $\sum |\Delta\alpha_k| \leq V$.) By Theorem 7.2, $f \in \mathcal{R}(\alpha)$.

Step 2: Interchange of limit and integral.

For $n \geq N$ (with $|f_n(x) - f(x)| < \varepsilon/(V+1)$ for all x):

$$\begin{aligned} \left| \int_a^b f_n\,d\alpha - \int_a^b f\,d\alpha \right| &= \left| \int_a^b (f_n - f)\,d\alpha \right| \\ &\leq \sup_{x\in[a,b]} |f_n(x) - f(x)| \cdot V_\alpha(a,b) \\ &< \frac{\varepsilon}{V+1} \cdot V < \varepsilon. \end{aligned}$$

Hence $\int_a^b f_n\,d\alpha \to \int_a^b f\,d\alpha$. $\square$

4 Section D: Measure Theory

4.1 Definitions

Definition 4.1 (Sets of Measure Zero)**.** A set $E \subseteq \mathbb{R}$ has **(Lebesgue) measure zero** if for every $\varepsilon > 0$ there exists a countable collection of open intervals $\{I_n\}_{n=1}^{\infty}$ such that

$$E \subseteq \bigcup_{n=1}^{\infty} I_n \quad \text{and} \quad \sum_{n=1}^{\infty} \ell(I_n) < \varepsilon,$$

where $\ell(I_n)$ denotes the length of interval I_n.

Remark 4.1. Properties of sets of measure zero:

(i) Every countable set has measure zero.

(ii) A subset of a set of measure zero has measure zero.

(iii) A countable union of sets of measure zero has measure zero.

Definition 4.2 (Lebesgue Outer Measure)**.** For any set $E \subseteq \mathbb{R}$, the **Lebesgue outer measure** of E is defined by

$$m^*(E) := \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) : E \subseteq \bigcup_{n=1}^{\infty} I_n, \text{ each } I_n \text{ is an open interval} \right\}.$$

Definition 4.3 (Measurable Set (Lebesgue Measurable))**.** A set $E \subseteq \mathbb{R}$ is called **(Lebesgue) measurable** if for every set $A \subseteq \mathbb{R}$:

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c),$$

where $E^c = \mathbb{R} \setminus E$. This is known as the **Carathéodory criterion**.

Equivalently, E is measurable if it "splits" every test set A additively with respect to outer measure. The collection of all measurable sets forms a σ-algebra.

Definition 4.4 (Borel Sets)**.** The **Borel σ-algebra** $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra containing all open sets in $\mathbb{R}$ (equivalently, the smallest σ-algebra containing all open intervals). The elements of $\mathcal{B}(\mathbb{R})$ are called **Borel sets**.

The Borel σ-algebra can be generated by any of the following:

- All open sets.
- All closed sets.
- All open intervals (a, b).
- All half-open intervals $(-\infty, a]$, $a \in \mathbb{R}$.

Every Borel set is Lebesgue measurable, but not every Lebesgue measurable set is a Borel set.

Definition 4.5 (Measurable Function)**.** Let E be a measurable subset of $\mathbb{R}$. A function $f : E \to \mathbb{R} \cup \{-\infty, +\infty\}$ is called **(Lebesgue) measurable** if for every $\alpha \in \mathbb{R}$, the set

$$\{x \in E : f(x) > \alpha\}$$

is a measurable set.

Equivalently, any of the following conditions may replace the one above:

- $\{x \in E : f(x) \geq \alpha\}$ is measurable for all $\alpha \in \mathbb{R}$.
- $\{x \in E : f(x) < \alpha\}$ is measurable for all $\alpha \in \mathbb{R}$.
- $\{x \in E : f(x) \leq \alpha\}$ is measurable for all $\alpha \in \mathbb{R}$.
- $f^{-1}(G)$ is measurable for every open set $G \subseteq \mathbb{R}$.

Remark 4.2. Key properties of measurable functions:

(i) Every continuous function is measurable.

(ii) Every monotone function on an interval is measurable.

(iii) If f and g are measurable, then $f + g$, fg, $|f|$, $\max(f, g)$, $\min(f, g)$ are measurable.

(iv) If $\{f_n\}$ is a sequence of measurable functions, then $\sup_n f_n$, $\inf_n f_n$, $\limsup f_n$, $\liminf f_n$ are all measurable.

(v) The pointwise limit of a sequence of measurable functions (when it exists) is measurable.